

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459310.7091 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.057
Transit depth (Rp/Rs)^2	3.09 +/- 1.99 [%]
Orbital Inclination (inc)	79.02 +/- 2.78
Airmass coefficient 1 (a1)	0.24 +/- 0.15
Airmass coefficient 2 (a2)	1.07 +/- 1.37
Scatter in the residuals of the lightcurve fit is	69.55 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.035 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.176 ± 0.057 vs 0.1594 (deviation 0.25σ)
Duration fit vs published	0.035 d vs 0.0512 d (deviation 0.78σ)
Mid-time O-C	-0.8 min
Depth SNR	2.7
Residual scatter	69.55 %

Light curve

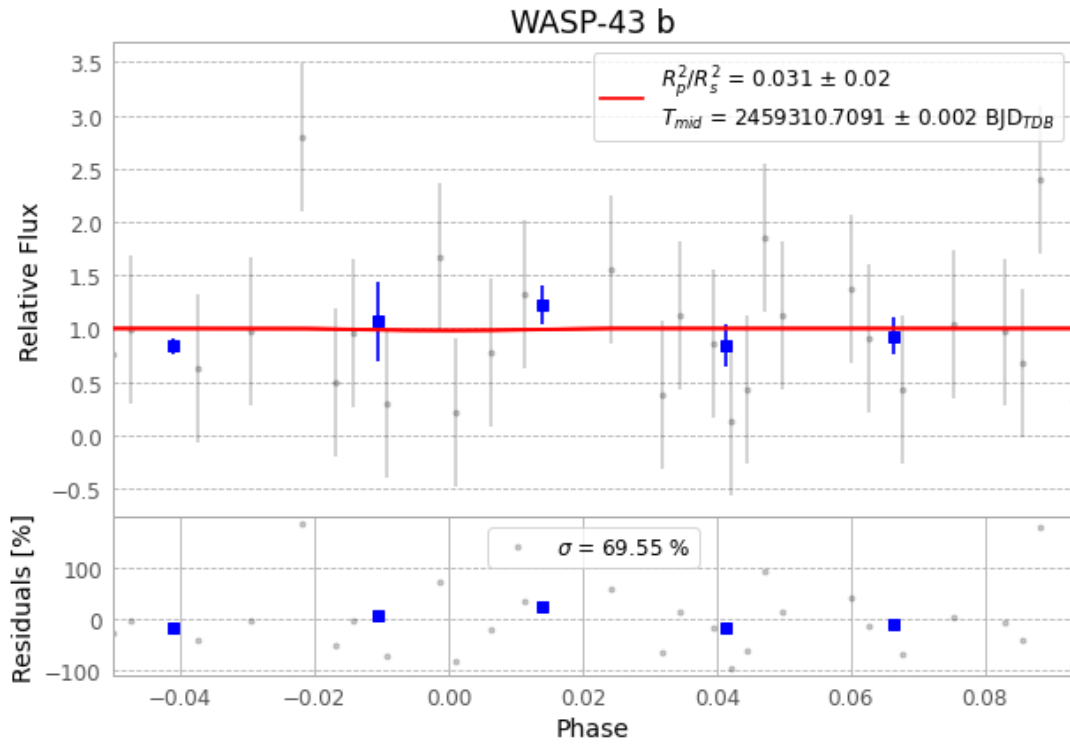


Figure 1 — FinalLightCurve_WASP-43 b_2021-04-05.png (SHA-256 5929c712d7003739...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [585, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 459fdf273c0c02d4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

58 FITS frames (38 MB), WASP-43210406034512.FITS → WASP-43210406064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210406.log (ecce0deafd0e...), FinalLightCurve_WASP-43 b_2021-04-05.png (5929c712d700...), FinalLightCurve_WASP-43 b_2021-04-05.csv (6fe50485865e...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 05:20Z.